

a b

$$z = 7 + 7i$$

$$r \operatorname{cis}(\theta)$$

$$r = \sqrt{a^2 + b^2} \quad \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

→ A polar

$$r = \sqrt{7^2 + 7^2} = \sqrt{65}$$

$$\theta = \tan^{-1}\left(\frac{7}{7}\right)$$

$$= 0.51$$

$$z = \sqrt{65} \cdot \operatorname{cis}(0.51)$$

Forma exponencial $r \cdot e^{i\theta}$

$$r = \sqrt{a^2 + b^2} \quad \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\sqrt{65} \cdot e^{i0.51}$$

Dependiendo de esta tabla lo de $\tan^{-1}\left(\frac{b}{a}\right)$ hay sumar o restar π

Cuadrante	a	b	Rango	Que hago?
I	+	+	Ya esta	NADA
II	-	+	Esta Fuera	$+\pi$ al θ
III	-	-	Esta Fuera	$-\pi$ al θ
IV	+	-	Ya esta	NADA

$$\theta = \frac{\pi}{2} \quad \text{si } a = 0 \quad \text{y } b > 0 \quad z = 3i$$

$$\theta = -\frac{\pi}{2} \quad \text{si } a = 0 \quad \text{y } b < 0 \quad z = -3i$$

$$\theta = \left[\tan^{-1}\left(\frac{b}{a}\right) \right] \pm \pi$$

Teorema de Moivre

z_1, z_2 , ambos en forma polar o exponencial

$$\left. \begin{aligned} r_1 \operatorname{cis}(\theta_1) \cdot r_2 \operatorname{cis}(\theta_2) \\ = r_1 \cdot r_2 \operatorname{cis}(\theta_1 + \theta_2) \end{aligned} \right\} \text{Polar}$$

$$\left. \begin{aligned} r_1 \cdot e^{i\theta_1} \cdot r_2 \cdot e^{i\theta_2} \\ = r_1 \cdot r_2 \cdot e^{i(\theta_1 + \theta_2)} \end{aligned} \right\} \text{Exponencial}$$

$\frac{z_1}{z_2}$ ambos en forma polar o exponencial

$$\left. \begin{aligned} \frac{r_1 \operatorname{cis}(\theta_1)}{r_2 \operatorname{cis}(\theta_2)} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2) \end{aligned} \right\} \text{Polar}$$

$$\left. \begin{aligned} \frac{r_1 \cdot e^{i\theta_1}}{r_2 \cdot e^{i\theta_2}} = \frac{r_1}{r_2} \cdot e^{i(\theta_1 - \theta_2)} \end{aligned} \right\} \text{Exponencial}$$

$$[r \operatorname{cis}(\theta)]^n = r^n \cdot \operatorname{cis}(\theta \cdot n) \text{ Polar}$$

$$[r e^{i\theta}]^n = r^n \cdot e^{i\theta \cdot n} \text{ Exponencial}$$

el r se eleva y el
 θ se multiplica

Expresa esto en forma rectangular

$$\frac{\overbrace{(1-i)}^{Z_1}{}^6 \overbrace{(1-i\sqrt{3})}^{Z_2}{}^3}{\underbrace{(2i-2\sqrt{3})}_{Z_3}{}^7}$$

Cuadrante	a	b	Rango	Que hago?
I	+	+	Ya esta	NADA
II	-	+	Esta Fuera	$+\pi$ al θ
III	-	-	Esta Fuera	$-\pi$ al θ
IV	+	-	Ya esta	NADA

$$\theta = \frac{\pi}{2} \quad \text{si } a=0 \text{ y } b>0 \quad z=bi$$

$$\theta = -\frac{\pi}{2} \quad \text{si } a=0 \text{ y } b<0 \quad z=-bi$$

$$\theta = \left[\tan^{-1}\left(\frac{b}{a}\right) \right] \pm \pi$$

1) Pasar todo a polar

2) Recordar sumar o restar π si es necesario

3) Aplicar exponete

$$r = \sqrt{a^2 + b^2} \quad \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$Z_1 = 1 - i$$

$$r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \arctan\left(\frac{-1}{1}\right) = -\frac{\pi}{4}$$

$$Z_1 = \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)$$

$$r \operatorname{cis}(\theta)$$

$$Z_2 = 1 - i\sqrt{3}$$

$$r = \sqrt{1^2 + (-\sqrt{3})^2} = 2$$

$$\theta = \tan^{-1}\left(\frac{-\sqrt{3}}{1}\right) = -\frac{\pi}{3}$$

$$Z_2 = 2 \cdot \operatorname{cis}\left(-\frac{\pi}{3}\right)$$

$$r \operatorname{cis}(\theta)$$

$$Z = 2i - 2\sqrt{3}$$

$$= -2\sqrt{3} + 2i$$

$$r = \sqrt{(-2\sqrt{3})^2 + 2^2} = 4$$

$$\theta = \arctan\left(\frac{2}{-2\sqrt{3}}\right) + \pi = \frac{5\pi}{6}$$

$$Z_3 = 4 \cdot \operatorname{cis}\left(\frac{5\pi}{6}\right)$$

$$\frac{\overbrace{(1-i)}^{Z_1}{}^6 \overbrace{(1-i\sqrt{3})}^{Z_2}{}^3}{\underbrace{(2i-2\sqrt{3})}_{Z_3}{}^7}$$

$$Z_1 = \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)$$

$$Z_2 = 2 \cdot \operatorname{cis}\left(-\frac{\pi}{3}\right)$$

$$Z_3 = 4 \cdot \operatorname{cis}\left(\frac{5\pi}{6}\right)$$

$$\frac{(\sqrt{2} \operatorname{cis}(-\frac{\pi}{4}))^6 (2 \cdot \operatorname{cis}(-\frac{\pi}{3}))^3}{(4 \cdot \operatorname{cis}(\frac{5\pi}{6}))^7}$$

$$\frac{(\sqrt{2} \operatorname{cis}(-\frac{\pi}{4}))^6 (2 \operatorname{cis}(-\frac{\pi}{3}))^3}{(4 \operatorname{cis}(\frac{5\pi}{6}))^7}$$

$$[r \operatorname{cis}(\theta)]^n = r^n \cdot \operatorname{cis}(\theta \cdot n) \text{ Polar}$$

$$\frac{(\sqrt{2})^6 \cdot \operatorname{cis}(-\frac{\pi}{4} \cdot 6) \cdot 2^3 \cdot \operatorname{cis}(-\frac{\pi}{3} \cdot 3)}{4^7 \cdot \operatorname{cis}(\frac{5\pi}{6} \cdot 7)}$$

$$\frac{r_1 \operatorname{cis}(\theta_1)}{r_2 \operatorname{cis}(\theta_2)} = \frac{r_1}{r_2} \cdot \operatorname{cis}(\theta_1 - \theta_2) \text{ Polar}$$

$$\frac{2^3 \cdot 2^3}{4^7 \rightarrow 2^{14}} \cdot \frac{\operatorname{cis}(-\frac{6\pi}{4}) \cdot \operatorname{cis}(-\pi)}{\operatorname{cis}(\frac{35\pi}{6})}$$

$$\left. \begin{array}{l} r_1 \operatorname{cis}(\theta_1) \cdot r_2 \operatorname{cis}(\theta_2) \\ r_1 \cdot r_2 \operatorname{cis}(\theta_1 + \theta_2) \end{array} \right\} \text{ Polar}$$

$$\frac{2^3 \cdot 2^3}{2^{14}} \cdot \frac{\operatorname{cis}(-\frac{6\pi}{4} + -\pi)}{\operatorname{cis}(\frac{35\pi}{6})}$$

$$\frac{2^3 \cdot 2^3}{2^{14}} \cdot \frac{\operatorname{cis}(-\frac{5\pi}{2})}{\operatorname{cis}(\frac{35\pi}{6})}$$

$$\frac{r_1 \operatorname{cis}(\theta_1)}{r_2 \operatorname{cis}(\theta_2)} = \frac{r_1}{r_2} \cdot \operatorname{cis}(\theta_1 - \theta_2) \text{ Polar}$$

$$\frac{2^3 \cdot 2^3}{2^{14}} \cdot \operatorname{cis}\left(\frac{-5\pi}{2} - \frac{35\pi}{6}\right)$$

$$\frac{2^3 \cdot 2^3}{2^{17}} \cdot \left(\cos\left(\frac{-5\pi}{2} - \frac{35\pi}{6}\right) \right)$$

$$\frac{1}{2^8} \cdot \left(\cos\left(\frac{-25\pi}{3}\right) \right) \quad \begin{matrix} \text{Polar} \\ \rightarrow \text{Rectangular} \end{matrix}$$

$$\frac{1}{2^8} \cdot \left[\cos\left(\frac{-25\pi}{3}\right) + i \sin\left(\frac{-25\pi}{3}\right) \right]$$

$$\boxed{\frac{1}{2^8} \cos\left(\frac{-25\pi}{3}\right) + \frac{1}{2^8} \sin\left(\frac{-25\pi}{3}\right) i}$$

$$i^0 = 1$$

$$i^1 = i \quad i^n = n - \left[\text{int}\left(\frac{n}{4}\right) \cdot 4 \right]$$

$$i^2 = -1$$

$$i^3 = -i$$

$$i^{100} = 100 - \left(\frac{100}{4}\right) \cdot 4 = 100 - 100 = 0$$

$$\begin{matrix} i^0 = 1 \\ i^{100} = 1 \end{matrix}$$

$$i^{65} = 65 - \text{int}\left(\frac{65}{4}\right) \cdot 4$$

$$65 - 16 \cdot 4 = 1 \rightarrow i^1 = i \rightarrow i^{65} = i$$

$$i^{30} = 30 - \text{int}\left(\frac{30}{4}\right) \cdot 4$$

$$30 - 7 \cdot 4 = 2 \rightarrow i^2 = -1 \rightarrow i^{30} = -1$$

$$i^{19} = 19 - \text{int}\left(\frac{19}{4}\right) \cdot 4$$

$$19 - 4 \cdot 4 = 3 \rightarrow i^3 = -i \rightarrow i^{19} = -i$$

$$\frac{\overbrace{(3-7i)}^{z_1}}{7i} z_2$$

$$\begin{array}{cc} a & b \\ - & + \\ - & - \end{array}$$

$$z_1 = \overset{a}{3} - \overset{b}{7}i$$

$$r = \sqrt{3^2 + (-7)^2} = 5$$

$$\theta = \arctan\left(\frac{-7}{3}\right) = -0,92$$

$$z_1 = 5 \cdot \text{cis}(-0,92)$$

$$z_2 = 0 + 7i$$

$$r = \sqrt{0^2 + 7^2} = 7$$

$$\theta = \frac{\pi}{2}$$

$$z_2 = 7 \cdot \text{cis}\left(\frac{\pi}{2}\right)$$

$$\frac{5 \cdot \text{cis}(-0,92)}{7 \cdot \text{cis}\left(\frac{\pi}{2}\right)}$$

$$\frac{5}{7} \cdot \text{cis}\left(-0,92 - \frac{\pi}{2}\right)$$

$$\frac{5}{7} \left[\cos\left(-0,92 - \frac{\pi}{2}\right) + i \sin\left(-0,92 - \frac{\pi}{2}\right) \right]$$

$$\frac{5}{7} \cdot \cos\left(-0,92 - \frac{\pi}{2}\right) + \frac{5}{7} \sin\left(-0,92 - \frac{\pi}{2}\right)i$$

Logaritmo principal

$$\ln^2 \left(\frac{i^{2027}}{1+i} \right) \quad \ln^2(x) = [\ln(x)]^2$$

$$\left[\ln \left(\frac{\overbrace{i^{2027}}^{z1}}{\underbrace{1+i}_{z2}} \right) \right]^2$$

Hay que pasar
todo a exponencial
 $r \cdot e^{i\theta}$

$$i^{2027} = 2027 - \text{int} \left(\frac{2027}{4} \right) \cdot 4$$

$$2027 - 506 \cdot 4 = 3$$

$$i^{2027} = i^3 = -i \rightarrow z1 = 1 + 0i$$

$$\begin{matrix} a & b \\ z2 = 1 + i \end{matrix}$$

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

$$z2 = \sqrt{2} \cdot e^{i \cdot \frac{\pi}{4}}$$

$$\begin{matrix} a & b \\ z1 = 1 + 0i \end{matrix}$$

$$r = \sqrt{1^2 + 0^2} = 1$$

$$\theta = \arctan\left(\frac{0}{1}\right) = 0$$

$$z1 = 1 \cdot e^{i0}$$

$$\left[\ln \left(\frac{i^{2027}}{1+i} \right) \right]^2$$

$$\left[\ln \left(\frac{1 \cdot e^{i0}}{\sqrt{2} \cdot e^{i \frac{\pi}{4}}} \right) \right]^2$$

$$\left[\ln \left(\frac{1}{\sqrt{2}} \cdot e^{i(0 - \frac{\pi}{4})} \right) \right]^2$$

$$\left[\ln \left(\frac{1}{\sqrt{2}} \cdot e^{-\frac{\pi}{4}i} \right) \right]^2$$

$$\ln(a \cdot b) = \ln(a) + \ln(b)$$

$$\left[\ln \left(\frac{1}{\sqrt{2}} \cdot e^{-\frac{\pi}{4}i} \right) \right]^2$$

$$\left[\ln \left(\frac{1}{\sqrt{2}} \right) + \cancel{\ln(e)}^{-\frac{\pi}{4}i} \right]^2$$

$$\left[\ln \left(\frac{1}{\sqrt{2}} \right) - \frac{\pi}{4}i \right]^2 \quad \begin{matrix} (a-b)^2 \\ a^2 - 2ab + b^2 \end{matrix}$$

$$\ln^2 \left(\frac{1}{\sqrt{2}} \right) - 2 \ln \left(\frac{1}{\sqrt{2}} \right) \cdot \frac{\pi}{4}i + \left(\frac{-\pi i}{4} \right)^2$$

$$\ln^2 \left(\frac{1}{\sqrt{2}} \right) - 2 \ln \left(\frac{1}{\sqrt{2}} \right) \cdot \frac{\pi}{4}i - \frac{\pi^2}{16} \quad i^2 = -1$$

$$\left[\ln^2 \left(\frac{1}{\sqrt{2}} \right) - \frac{\pi^2}{16} \right] + \left[-2 \ln \left(\frac{1}{\sqrt{2}} \right) \cdot \frac{\pi}{4} \right] i$$